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Carbon Emission Tax Policy in an Intermodal Transportation Network

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Abstract

In this paper, we empirically studied and proposed a carbon emission tax policy based on a real world coal transportation intermodal network across 15 states in the US including highway, railway and inland waterway. Theoretical analysis on boundary models motivates the potential improvement on environment while not increasing transportation cost. We develop multi-objective optimization models to analyze intermodal transportation with economic, time performance and environmental considerations. The computational study validates the environment improvement and provides authorities a potential carbon emission tax policy and a change in the system-wide transportation cost. The breakeven point for tax rate is 13 dollars per ton, which can provide minimum carbon emission without increasing the transportation cost.

Keywords
Intermodal transportation, network design, multi-objective optimization, carbon emission

1. Introduction
Transportation cost and time performance are always the main concerns in most logistic systems. Freight shipped only using highway account for about 34.6 % in the US [1]. However, at the same time, carbon dioxide emission caused by freight road transportation has increased rapidly due to long travel distance. According to the US Environmental Protection Agency (EPA), around 28% of all carbon dioxide emissions were due to transportation which is the second largest contributor in the US. Among these, over two-thirds come from road freight transportation. Consequently, there is a need to design an efficient and environmentally friendly logistical system to satisfy both government and carriers. However, for-profit companies and governments may have conflicting objectives relative to freight transportation. To minimize the environmental impact, government agencies intend to develop policies that shift freight transportation to other transportation modes such as railway and waterway. Companies typically pursue fast, minimum cost transportation strategies while satisfying the limitations of the regulatory agencies. Changes on distribution modes would potentially lead to different market segmentation and further impact on pricing decisions [2].

The combination of operational decisions in supply chain and the environmental issues have been studied in several aspects of logistics: inventory and facility decisions, design, planning and control for transportation [3]. Environmental impacts have been internalized in many papers in the green supply chain management literature. Articles in this field deal with how much to produce or order from one or more sources taking into account environmental impacts, which can range from carbon emissions to waste. Benjaafar et al. [4] combined operational decisions in supply chain and the environmental issues together by integrating carbon emission parameters into traditional operations models. They evaluate how regulatory emission control policies affect cost and emissions. Elhedhli and Merrick [5] considered emission costs associated with fixed cost, variable location and production costs in their distribution network model. Their result shows that rigidity of the problem has a large impact on the load ratio of the open distribution centers. Tang et al. [6] integrate environmental considerations in the classical incapacitated facility location problem which aims to minimize economic cost and CO_2 emissions while providing strategic facility locations within a logistics network. Hoen et al. [7] focus on reducing carbon emission costs by switching transport modes within an existing network. Konur and Schaefer [8] proposed Carbon emissions regulation policies by studying EOQ model with less-than-truckload and truckload carriers.

A majority of the green supply chain literature takes a carrier company's perspective by focusing on problems with a single transportation mode and a single origin-destination (OD) pair. Palmer [9] developed a vehicle routing model that calculates total carbon emissions, transportation time and travel distance to look for more environmentally beneficial routes. This model provided a delivery strategy corresponding to various carbon emission regulations. His work is extended by Maden et al. [10]. Bektas and Laporte [11] investigate the effect of load and speed of truck on carbon emissions in their Pollution-Routing problem using a comprehensive emission model. Another important topic with respect to transportation is the transportation mode selection. Bloemhof et al. [12] investigate energy use and emissions for different modes and compare the environmental impacts of inland navigation to rail and road transport. Vanek and Morlok [13] focus on improving the energy efficiency of frequently used trucks, taking into account shifting cargo to rail transportation. Mallidis et al. [14] introduce port entry choice in a supply chain model because it is known that waterway is an energy-efficient transportation method.

Incorporation of multi-commodity intermodal transportation and carbon emissions in supply chain design is almost absent in the literature. However, it is becoming more critical to integrate environmental issues and intermodal transportation as most papers suggest. In this paper, we adopt the viewpoint of a government agency and focus on an intermodal transportation system consisting of three modes: inland waterway, railway and highway. The intermodal transportation network design problem has multiple objectives. Boundary models are formulated to evaluate the potential improvement of the network. In particular, we develop a railway, waterway and highway intermodal network using ArcGIS. The demand data were obtained from the Bureau of Transportation Statistics (BTS) under the US DOT. Based on the analysis of the current transportation pattern choice, network flows and corresponding carbon emission data, a multi-objective optimization model is proposed to simulate the commodity flows on railways, waterways and highways for the nation's coal transportation system across the fifteen states. Carbon tax is used as an emission regulation parameter to adjust the trade-off between economic cost and carbon emission.

The remainder of the paper is organized as follows. In Section 2, four boundary models and a multi-objective intermodal transportation model are described. In Section 3, we conduct a scenario analysis on a real-life case study. We end with the conclusion and discussions in Section 4.

2. Intermodal Transportation Models

2.1 Definitions and Notations

Consider a network $G = (N, A)$, where N is the set of nodes and A is the set of arcs that includes waterway, highway and railway. Given a fixed demand, we not only intend to optimize the transportation mode for each company, but also aim to reduce the overall carbon emissions for the entire network. The optimization constraints are: (a) material flow demand constraints, (b) the flow balance constraints for each transportation mode, (c) the freight system capacity constraints, (d) the carbon emission restriction constraints (carbon tax (beta) is used as the regulation parameter for emissions), (e) the non-negativity constraints, and (f) the time penalty constraints. Note that the unit time penalty cost is defined as α which is used to simulate the actual transportation cost.

Notations used in models are as follows.
f_{OD}: Demand from an origin O to a destination D, $O \in N$, $D \in N$, $O \neq D$
M: Set of transportation modes
c^m: Unit cost per mile per ton using transportation mode m
D_{ijm}: Distance between node i and j using transportation mode m, $i \neq j$
Cap_{ijm}: Total capacity between node i and j using transportation mode m, $i \neq j$, $i = 1,2 \dots N$, $j = 1,2 \dots N$
$h_{ijm} = \begin{cases} 1, & Cap_{ijm} > 0 \\ 0, & otherwise \end{cases}$
v_m: Average travel speed for mode m
x_{ijm}^{OD}: Material flow from node i to node j using transportation mode m for demand f_{OD}, $i \neq j$
α: Unit time penalty cost
β: Carbon tax (dollar per ton)

2.2 Boundary Models

We first consider a typical minimum cost problem, where the total transportation cost is minimized while the flow balance and capacity restrictions are satisfied. The model is formulated as (1) – (6). Model 1 would provide a lower bound of the transportation cost in the network, and thus help to evaluate the efficiency of current logistics performance.

Model 1: Min. $\sum_{(O,D)} \sum_{i\in N} \sum_{j\in N} \sum_{m\in M} c^m D_{ijm} x_{ijm}^{OD}$ (1)

S.T. $\sum_{j\in N} \sum_{m\in M} x_{Ojm}^{OD} = f_{OD},\ \ if\ j \neq O, \forall (O,D)$ (2)

$\sum_{i\in N} \sum_{m\in M} x_{iDm}^{OD} = f_{OD},\ \ if\ i \neq D, \forall (O,D)$ (3)

$\sum_{j\in N} \sum_{m\in M} x_{jim}^{OD} = \sum_{j\in N} \sum_{m\in M} x_{ijm}^{OD},\ if\ i \neq O\ and\ i \neq D, i \in N, i \neq j, \forall (O,D)$ (4)

$\sum_{(O,D)} x_{ijm}^{OD} \leq Cap_{ijm}, i \in N, j \in N, i \neq j, m \in M$ (5)

$x_{ijm}^{OD} \geq 0, i \in N, j \in N, i \neq j$ (6)

The objective function calculates the total transportation cost, which is impacted by the arcs material travels on and the transportation model adopted. Constraint (2) requires that the total outgoing flow from an origin to a destination is equal to the demand of this O-D pair. Constraint (3) requires that the total incoming flow to a destination port equals the fixed demand for this O-D pair. Constraint (4) guarantees the balance of the incoming and outgoing flows at each transfer node for each O-D pair demand. Note that these flow balance constraints should be defined based on each O-D pair respectively. Different O-D pairs cannot be aggregated at a node, because the material flow of an O-D pair coming to a node cannot be mixed with material flow from another O-D pair. Constraint (5) restricts that the total material flow on an arc cannot exceed the capacity for the particular transportation mode. Constraint (6) guarantees the non-negativity of the flow variables.

Based on the optimal solution of Model 1, transportation cost C_1 is obtained directly from the objective value, and carbon emission volume E_1 can be calculated as $E_1 = \sum_{i\in N} \sum_{j\in N} \sum_{m\in M} f^E\ (D_{ijm}, \sum_{(O,D)} x_{ijm}^{OD})$, where the f^E is the carbon emission function, which will be explored further in section 3.

Focusing on cost reduction could lead to increase in carbon emission. Thus, a constraint based on current carbon emission volume will help obtain a less-aggressive lower bound on transportation cost. Constraint (7) restricts the total emission of an optimal solution cannot exceed the current volume, E_0. The total transportation cost and carbon emission at the optimum of Model 2 are C_2 and E_2 respectively.

Model 2: Min. $Objective\ (1)$

S.T. $\sum_{i\in N} \sum_{j\in N} \sum_{m\in M} f^E\ (D_{ijm}, \sum_{(O,D)} x_{ijm}^{OD}) \leq E_0$ (7)

$Constraints\ (2) - (6)$

From another aspect, we are also interested in how best the carbon emission can be controlled. Model 3 minimizes the total emission with flow balance and capacity restrictions the same as Model 1. At the optimum, the objective value is the carbon emission E_3 and total transportation cost C_3 can be calculated based on flows as $C_3 = \sum_{(O,D)} \sum_{i\in N} \sum_{j\in N} \sum_{m\in M} c^m D_{ijm} x_{ijm}^{OD}$.

Model 3: Min. $\sum_{i\in N} \sum_{j\in N} \sum_{m\in M} f^E\ (D_{ijm}, \sum_{(O,D)} x_{ijm}^{OD})$ (8)

S.T. $Constraints\ (2) - (6)$

It is possible that Model 3 achieves an optimum with unrealistically low carbon emission. Model 4 minimizes the total emission with an additional restriction of the current cost, C_0, to achieve a less-aggressive lower bound for carbon emission. The total transportation cost and carbon emission at the optimum of Model 4 are C_4 and E_4 respectively.

Model 4: Min. $Objective\ (8)$

S.T. $\sum_{(O,D)} \sum_{i\in N} \sum_{j\in N} \sum_{m\in M} C_{ijm} x_{ijm}^{OD} \leq C_0$ (9)

$Constraints\ (2) - (6)$

In summary, the optimal solutions of the above four models have relationships as
$C_1 \leq C_2 \leq C_0 \geq C_4 \leq C_3$ and $E_1 \geq E_2 \leq E_0 \geq E_4 \geq E_3$.

The gaps between current carbon emission and bound values indicate the potential improvement one can obtain by adjusting the material flows. Along with relationships between corresponding transportation costs, it is realized that opportunities may exist for reducing cost and emission at the same time. Detailed numerical analysis is conducted in section 4. Because the current transportation flow is not optimal, a policy is studied in this paper to force people to reconstruct flows to reduce the economic and environmental costs simultaneously.

2.3 Multi-objective Model

To control the carbon emission associated with transportation, one of multiple feasible solutions is to collect the pollution tax. Before studying the impact of tax rate, we first need to understand and model people's behavior in the current situation. Based on current government policy today, most people usually are not concerned about pollution when deciding a route choice. However, the delivery time has an impact on routing and drives the resulting route from the distance based optimum. With an assumed parameter α as the time penalty, Model 5 simulates people's choice in the transportation network.

Model 5: Min. $\sum_{(O,D)}\sum_{i\in N}\sum_{j\in N}\sum_{m\in M} c^m D_{ijm} x_{ijm}^{OD} + \alpha \sum_{(O,D)}\sum_{i\in N}\sum_{j\in N}\sum_{m\in M}\left(\frac{D_{ijm}}{v_m} - \min_m\left\{\frac{D_{ijm}}{h_{ijm}v_m}\right\}\right) x_{ijm}^{OD}$ (10)

S.T. $Constraints\ (2)-(6)$

Objective (10) minimizes the distance based transportation cost and the time cost simultaneously. If people do not choose the fastest way on an arc, there is extra time penalty cost generated. Even though different individuals may have different time preferences, a unique α value represents an average level to simulate the behavior of an entire group. The proper α value can be obtained by comparing the actual transportation cost and the resulting value of the first term in the objective (10) at optimum. When the objective value is equal to C_0, the resulting transportation flow is treated as the actual traffic condition. And the resulting α is used for further analysis of carbon emission parameter.

With a fixed unit time penalty, the impact of pollution policy can be evaluated based on Model 6. Assume that tax is linear on the carbon emission volume with a coefficient β. Objective (11) in Model 6 additionally considers the environmental cost that people have to pay, and aims to minimize it with transportation as well as time costs.

Model 6: Min. $\sum_{(O,D)}\sum_{i\in N}\sum_{j\in N}\sum_{m\in M} c^m D_{ijm} x_{ijm}^{OD} + \alpha \sum_{(O,D)}\sum_{i\in N}\sum_{j\in N}\sum_{m\in M}\left(\frac{D_{ijm}}{v_m} - \min_m\left\{\frac{D_{ijm}}{h_{ijm}v_m}\right\}\right) x_{ijm}^{OD} + \beta \sum_{i\in N}\sum_{j\in N}\sum_{m\in M} f^E(D_{ijm}, \sum_{(O,D)} x_{ijm}^{OD})$ (11)

S.T. $Constraints\ (2)-(6)$

The total transportation cost and carbon emission at the optimum of Model 6 are C_6 and E_6 respectively. By adjusting β, a tax policy based on β can be obtained with an expectation that $E_6 < E_0$ and $C_6 + \beta E_6 \le \gamma C_0$, where γ is a subjective tolerance of the decision maker and $\gamma \ge 1$. Detailed numerical analysis is conducted in section 3.

3. Computational Results

3.1 Coal Transportation

In this section, we present an intermodal network including the entire inland waterway systems across 15 states in US (Minnesota, Wisconsin, Iowa, Illinois, Missouri, Indiana, Ohio, Pennsylvania, West Virginia, Kentucky, Arkansas, Tennessee, Mississippi, Alabama, and Louisiana). Our study is only focused on the shipment of coal. According to the Department of Transportation (DOT) statistics, the primary source of our data, the annual transport volume within these 15 states is approximately 85,444 Ktons, 4.3 % of these volumes transported through highway, 15.1 % transported through waterway, 53.1 % shipped using railway and 27.5 % using multi-modal transportation. Note that the network model required detailed information on distance of highway, waterway and railways. Geographical information systems (GIS) provided an efficient way to estimate these parameters. We used ESRI's ArcGIS 9.3 software to develop a GIS-based intermodal network which consists of three transportation modes. Figures 2 and 3 depict a subset of this network within 15 states since we only focus on OD pairs within these states. The red, blue and green lines represent railway, waterway and highway, respectively. The major interstate highways are: I10, I12, I22, I24, I26, I29, I35, I39, I40, I43, I44, I49, I55, I57, I59, I64, I65, I68-I86, I88, I90 and I99. The major railways are BNSF, CSX, Norfolk Southern, and Union Pacific Railroad.

The map provided by ArcGIS software is used as our primary network basis. In the original form, this network is not suitable for real world coal transportation since only a few origin and destination points are exactly intermodal nodes. Thus, we represent the origin and destination nodes by projecting them onto the closest intermodal nodes. Consequently, the intermodal nodes in our model include origin and destination points. Each existing highway, railway and waterway between a node pair is represented as an arc. The resulting network model of coal transportation in 15 states has 76 nodes and 8,451 arcs.

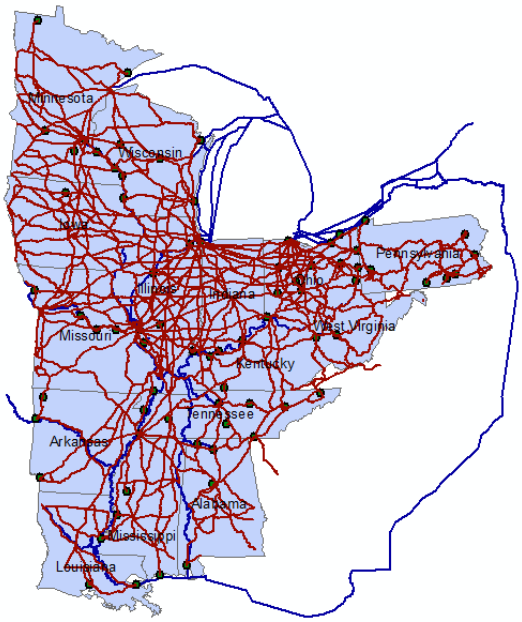


Figure 2: Railway and waterway network

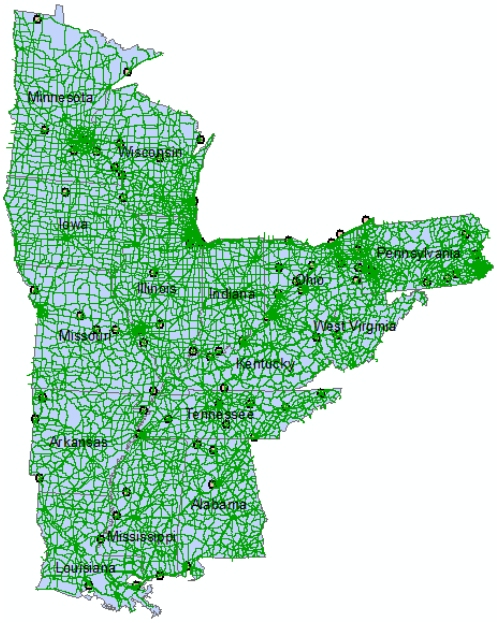


Figure 3: Highway network

The origins and destinations, as well as the demand of each OD pair are obtained from DOT. Specifically, demand is formed by the U.S. Department of Transportation OD matrix which contains coal transportation forecast for 2015. The regions defined by DOT are used as our origins and destinations. But in our work, we aggregate demand data for some regions and assign them to the nearest intermodal node in our ArcGIS model, as mentioned previously. The OD pairs that are assigned to intermodal nodes would potentially make use of the inland waterway system.

To obtain the actual cost, we analyze the transportation data from region to region of coal transportation within these fifteen states for 2015. The percentages of coal flowing from one region to another using different transportation mode are used to estimate the actual flow for year 2015. The resulting transportation cost is 450,834,574.9 dollars. We assume that the unit transportation cost is proportional to distance. According to the literature, one gallon of fuel can ship one ton of cargo 180-240 miles by highway, 450 miles by railway, and 514 miles by waterway (Wikipedia.org). Suppose one gallon fuel costs 4 dollars. Thus, the unit cost of truck is 0.02 dollars per ton per mile, unit cost of railway is 0.01 per ton per mile, and unit cost of waterway is 0.008 per ton per mile. For waterway, the capacity between two nodes is assumed to be the minimum capacity of these two nodes and locks between them. For highway, a full-width freeway lane typically has a capacity of 2,000 cars per hour (Wikipedia.org), and one car can carry 25 tons. We assume each highway has the same capacity. The annual highway capacity between two nodes can now be calculated as follows:

Cap_{ij2} =2,000*25*24*365/1000=438,000Ktons

For railway, a modern railcar has a gross capacity of 286,000 lbs or 125.5 tons moving in trains consisting of 100 cars or more, yielding a total carrying capacity of 12,500 tons. Suppose there is one train between two nodes per day, the capacity between these two nodes is calculated as: $\text{Cap}_{ij3} = 12{,}500 * 365/1000 = 4{,}562.5\text{ktons}$

According to the actual coal transportation data of 15 states in US, we analyze the traffic flow of different transportation modes (the percentage of each mode between states). The current transportation cost, $C_0 = 454{,}322{,}410$ dollars is obtained by multiplying the unit cost. The carbon emission volume, $E_0 = 915{,}957$ tons, are calculated based on model 6. Because we only consider coal transportation in our study, all vehicles are assumed to be fully loaded. Transshipment costs and external costs are not taken into account.

3.2 Carbon Emission Functions

The carbon emission functions for three different transportation modes are derived in this section respectively [15]. The average speed for each mode is taken account to estimate the emission function because we only consider annual demands.

For highway, we assume there is only one vehicle type: diesel truck without catalysts. Because a full truckload truck can carry 25 tons and the average speed of a full truckload is assumed to be 47 miles per hour (75.6km/h) (including traffic jams or queues at intersections), carbon emission for road can be calculated as follows:

$$E_{ij}^{hot} = 765 - 7.04\mathrm{V}_{ij} + 0.000632\mathrm{V}_{ij}^3 + \frac{8334}{\mathrm{V}_{ij}}$$

$$= 765 - 7.04 * 75.6 + 0.000632 * 75.6^3 + \frac{8334}{75.6} = 616.0894$$

Where

$$\varepsilon_i = 1.27 - \frac{0.483}{\mathrm{V}_i} = 1.27 - \frac{0.483}{75.6} = 1.2636$$

$$E_j^{start} = \omega_j \times \left(f_j(V) + g_j(T) - 1\right) \times h_j(d)$$

$$= 182.57 * (-0.0458 * 25 + 1.9163) * \frac{1 - e^{-\frac{3.95d}{0.24*75.6+0.09}}}{1 - e^{-3.95}}$$

$$= 143.58 * (1 - e^{-0.22d})$$

$$f^E = \sum_{i=1}^{I}\sum_{j=1}^{J}\left(E_{ij}^{hot}\varepsilon_{ij} + E_{ij}^{start} + E_{ij}^{evaporative}\right)NH_{ij}LH_{ij}$$

$$= \left(31.14 + 5.74 * \left(1 - e^{-0.22D_{ijm}}\right)\right) * \sum_{(O,D)} x_{ijm}^{OD} * D_{ijm}$$, for m=1 (truck mode)

For railway, some assumptions are made here to simplify the problem.(1) The average speed of train is 40 miles/h; (2) The mean distance between stops is 100 km; (3) the brake specific emission factor (BSEF) for CO_2 is assumed to be 42 g/kWh (PAN et.al [16]). Carbon emission of railway can thus be calculated as:

$$f^E = \sum_{i=1}^{M} 0.0036WSEC_i \times \frac{Tkm_i}{Tpt_i} \times BSEF_i \times LR_i$$

$$= 0.0036 * \left(0.019 * \frac{64.4^2}{\ln(64.4)} + 63\right) * \frac{\sum_{(O,D)} x_{ijm}^{OD}}{0.73} * 42 * D_{ijm}$$

$$= \frac{0.684}{\ln(100)} * \frac{\sum_{(O,D)} x_{ijm}^{OD}}{0.73} * 42 * D_{ijm} + 0.2268 * \frac{\sum_{(O,D)} x_{ijm}^{OD}}{0.73} * 42 * D_{ijm}$$

$$= 17 \sum_{(O,D)} x_{ijm}^{OD} * D_{ijm}$$, for m=2 (rail mode)

For waterway, the cargo capacity (maximum load) of a general cargo vessel for inland waterways is 3,840 ton. The fuel emission of CO_2 is 3,200 kg/ton. The average speed of a general cargo ship is 14.29 knots (26.5 km/h). Carbon emission for waterway can be calculated as follows:

$$f^E = \sum_{i=1}^{M} S_i F_i NW_i \frac{LW_i}{24V_i}$$

$$= 0.8 * (9.8197 + 0.00143 * 3840) * 3200 * 1000 * \frac{\sum_{(O,D)} x_{ijm}^{OD}}{3840} * \frac{D_{ijm}}{24 * 26.5}$$

$$= 16.05 \sum_{(O,D)} x_{ijm}^{OD} * D_{ijm}$$, for m=3 (river mode)

3.3 Optimal Intermodal Network

The optimal solution indicates the total transportation cost can be reduced to 340,295,206 dollars, and the corresponding emission is 727,183 tons. It turned out that the results of four boundary models are the same. More specifically, 30.4 percent cargo is transported using railway, 2.4 percent using highway and 67.2 percent using multi-modal transportation. This is the optimal solution of model 1 without considering carbon tax and time penalty cost. This is the ideal situation that the government agents aim to achieve. The reason for that is inland waterway transport is a more economical and environmentally friendly mode than railway and road transportation. In simpler terms, the government can reduce carbon emission by 188,774 tons if 67.2 percent of coal is transported using multi-modal transportation, whereas, carrier companies can save 114,027,205 dollars compared to the actual cost. The example of optimal route of model 1 is shown in Figure 4 and Table 2.

From Figure 4 and Table 2, it can be seen that, if the demand between Charleston to Louisville is 2,590 Ktons, the company can first ship 2,590 Ktons from Charleston to Huntington using railway; then ship 2,590 Ktons from Huntington to Cincinnati using waterway; finally transport these cargo from Cincinnati to Louisville using waterway. Similarly, if the demand between Columbus to Louisville is 896 Ktons, the resulting optimal route is: first using highway from Columbus to Cincinnati, then using waterway from Cincinnati to Louisville using waterway.

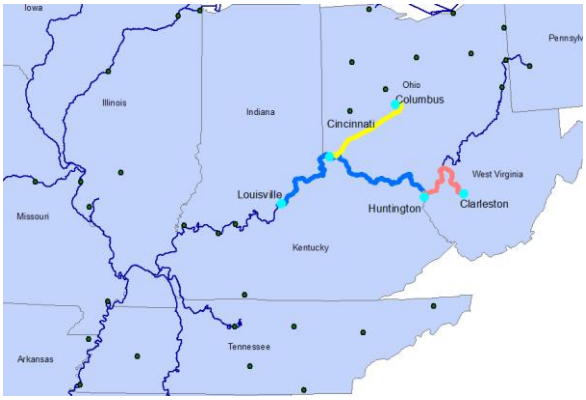


Figure 4: Model 1 optimal route snapshot

Table 2: Model 1 optimal solution associated with Figure 4

	Origin	Destination	Fix Demand (Ktons)	Intermodal Node		Mode	Value (Ktons)
x	Charleston	Louisville	2,589.978	Charleston	Huntington	R	2,589.978
				Huntington	Cincinnati	W	2,589.978
				Cincinnati	Louisville	W	2,589.978
x	Columbus	Louisville	895.507	Columbus	Cincinnati	H	895.507
				Cincinnati	Louisville	W	895.507

However, the reality is that many companies may not choose the optimal route due to various reasons. Among these, time performance is a critical one. In next section, first, we will use unit time penalty parameter to simulate the actual cost of carrier companies. Based on the estimated actual transportation routes, the trade-off between total costs (transportation cost and environmental cost) and carbon emission is presented. When carbon tax changes, the freight route choice and corresponding mode sharing in the intermodal system also change.

3.4 Carbon Emission Tax Policy

To find the best time penalty coefficient to simulate the real route choosing decision, we ran the optimization of Model 5 multiple times with different α values. As α changes, the model will choose different network flows and result in different transportation costs (the first term of objective (10) without the time penalty cost). It is showed that the increasing α value will drive the solutions away from the optimality of Model 1. When α equals to 1.65, the resulting transportation cost is the same as the actual total cost. Then, the optimal solution of Model 5 with $\alpha = 1.65$ is treated as the real network situation for the following tax policy analysis.

Model 6 is used to estimate the trade-off between payable cost (transportation cost and environmental cost) and carbon emission. We varied carbon tax rate from 0 to 14 based on the actual transportation route. With the introduction of carbon tax, the government can force the carriers to consider alternative transportation routes when shipping cargos.

The solutions of Model 6 with $\alpha = 1.65$ and different tax rates show that, as the tax rate β changes, network flows are selected differently. All the percentages are calculated comparing with the case of $\beta = 0$. Due to the properties of network, network flows will not rebuild until changes achieve a certain amount. For instance, when tax rate β increases from 0 to 1, there was a decrease in the overall rate of transportation cost (from 454,322,410.9 to 442,877,374.3 dollars), payable cost (from 454,322,410.9 to 443,777,894.3, 2.321%) and carbon emission (from 915,956 ton to 900,520 ton, 1.685%). Meanwhile, time penalty cost increase 9.923% compared to the base model 5.

However, when tax rate β increases from 1 to 11, there is no significant difference in transportation cost and carbon emission. Until carbon tax reaches 12, the transportation cost decreases from 442,877,374.3 to 442,460,839.4.

The payable cost does not exceed the current transportation cost until the carbon tax increases to 14 dollars per ton. Thus, a tax rate of 13 dollars per ton does not necessarily bring more costs to people if more routing optimization work is involved. At the same time, the carbon emission has been reduced by 1.755% directly, and government collects 11,698,492 dollars tax, which again can be used for environment improvement. The only disadvantage is the delivery time delay of 10.29%. This actually may not generate penalty cost, since an elastic margin of the delivery time in current real case always exists. In the case that people are sensitive on delivery times, a tax rate of 11 dollars per ton is a policy option, which has the same time delay as the case of 1 dollar per ton but would generate more tax revenue. On the other hand, if more time delay tolerance is allowed and people can accept an increase in total payable costs, the carbon tax rate can be more than 13 dollars per ton.

4. Conclusions

In this paper, we analyzed a real world intermodal network, which includes highway, railway and inland waterway and covers 15 states in US. Current transportation pattern choice, network flows and corresponding carbon emission data are collected and reported. With the classic carbon emission function, a set of mathematical programming models are built for further analysis. Four boundary models are built to evaluate how much an extra policy can potentially improve the network flows in terms of economic cost and environmental cost, and motivate us to seek a policy that can improve both simultaneously. The analytical results are also demonstrated by data based scenario analysis. An intermodal transportation model is thus built with objective to minimize the economic cost, time penalty and carbon emission together. The time penalty is introduced to simulate the real world group behavior through the mathematical model, and a proper penalty coefficient is found via numerical study. The sensitivity analysis of carbon emission tax policy shows that multiple tax rates apply for improving the economic transportation cost and controlling the carbon emission. The breakeven point for tax rate is 13 dollars per ton, which can bring maximum environment improvement while people do not pay more for transportation.

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